

EN401 — Handout 1: Energy Modelling Concepts & RES

Reference Energy System Architecture & Standardized Units

1. Technical Framework & Units Standardization

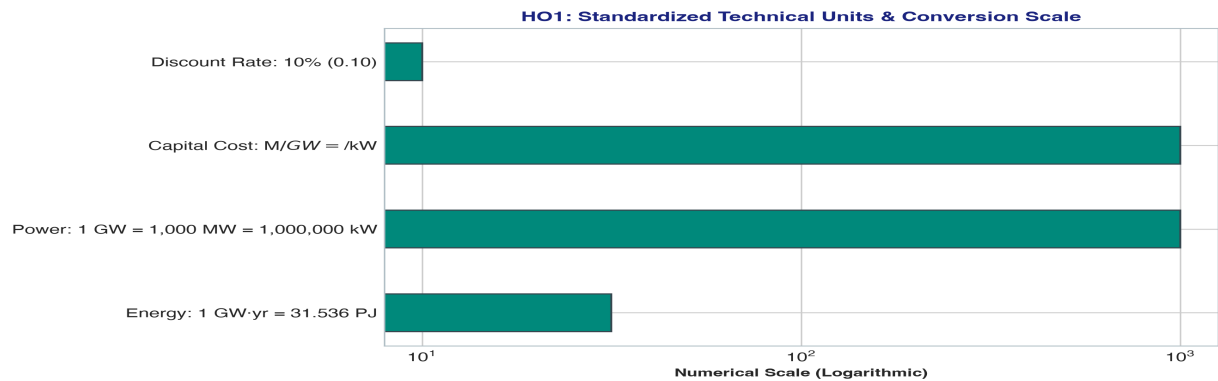


Figure: Technical units standardization framework in OSeMOSYS.

2. End-Use Electricity Demand Growth Trajectory (2021–2035)

Low-voltage electricity demand (ELC003) escalates linearly at +5.0 PJ/year from 20 PJ in 2021 to 90 PJ in 2035:

Year	Milestone	Demand (PJ/yr)	Electricity (TWh/yr)	Avg Power (GW)	Peak Load (GW)
2021	Horizon Start	20.00	5.56	0.634	0.762
2023	Early Phase	30.00	8.33	0.951	1.144
2025	Mid Checkpoint	40.00	11.11	1.268	1.525
2027	Expansion Era	50.00	13.89	1.586	1.906
2029	Expansion Era	60.00	16.67	1.903	2.287
2031	Late Phase	70.00	19.44	2.220	2.668
2033	Late Phase	80.00	22.22	2.537	3.050
2035	Horizon Final	90.00	25.00	2.854	3.431
Total	15-Yr Model	825.00	229.17	1.744	3.431 (Peak)

3. Core Takeaways

- Established standard units: Petajoules (PJ) for energy, Gigawatts (GW) for capacity.
- Formulated the deterministic LP minimization problem guaranteeing globally optimal solutions.